

Question	Answer	Marks							
4(a)(i)	sum of potential energy and kinetic energy	B1							
	(total) energy of random motion of particles	B1							
4(a)(ii)	potential energy (of molecules) (in an ideal gas) is zero, so the internal energy of the gas is equal to the total kinetic energy (of molecules)	B1							
	kinetic energy of molecules is proportional to (thermodynamic) temperature (so internal energy is proportional to (thermodynamic) temperature))	B1							
4(b)	cooling work done = 0	B1							
	compression increase in internal energy = $+2U$	B1							
	cooling change in internal energy = $-U$	B1							
	both rows: thermal energy adds to work to give increase in internal energy in terms of U and/or W (if fully correct, thermal energy for compression = $2U - W$ and thermal energy for cooling = $-U$: <table><tr><td>compression</td><td>$+W$</td><td>$2U - W$</td><td>$+2U$</td></tr><tr><td>cooling</td><td>0</td><td>$-U$</td><td>$-U$</td></tr></table>)	compression	$+W$	$2U - W$	$+2U$	cooling	0	$-U$	$-U$
compression	$+W$	$2U - W$	$+2U$						
cooling	0	$-U$	$-U$						

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